Indian Institute of Technology Jodhpur

PG Course Booklet

Indian Institute of Technology Jodhpur

July 2019

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Non Graded Institute Level Courses

For

M.Tech. and Ph.D.

Title	**Systems Engineering and Project Management**	Number	**OAL7XX0**
Department		L-T-P [C]	1-0-0 [1]
Offered for	M.Tech Program	Type	Compulsory NGU
Prerequisite			

Objectives

Systems Engineering is an interdisciplinary approach and means to enable realization of successful systems. It focuses on defining requirements and then proceeding with design synthesis and system validation while considering the complete problem including operations, performance, test, manufacturing, cost, and schedule.

This course intends to help students to develop the capability of systems thinking by introducing basics of systems engineering theory, methods, and tools through lecture and practice.

Learning Outcomes

After taking this class, you should be able to:

- Develop a systems engineering plan for a realistic project.
- Judge the applicability of any proposed process, strategy, or methodology for systems engineering
- Apply systems engineering tools (e.g., requirements development and management, robust design, Design Structure Matrix) to realistic problems.
- Know how to proactively design for and manage system lifecycle targets.
- Understanding TRL and MRL concepts
- Use project management tools

Contents

- Introduction - Terminology, Systems Thinking, Today's Challenges (1 L)
- Requirements Elicitation, Documenting System Requirements - Proper Specification Language, Depth of Definition (1 L)
- Requirements Analysis - Context Diagrams, Functional Analysis, State Diagrams (2 L)
- Requirements Validation - User Feedback Techniques, Requirements Reviews (1 L)
- System Architecture - Subsystem Definition, Hierarchical Analysis, Design for Reuse, Interface Management (2 L)
- Specialty Engineering - Producibility/Manufacturability, Reliability/Availability, Maintainability/Serviceability, Usability/User Interface, Design-to-Cost (2 L)
- Managing Systems Engineering - Estimating, Planning, Tracking, System Engineering

Management Plan, Work Breakdown Structure, Technical Reviews (2 L)

- Introduction to the concepts of TRL (Technology Readiness Levels) and MRL(Manufacturability Readiness level) (1 L)
- Course Project (8 hrs of self-work corresponding to 2 scheduled contact hours)
    - Application of Systems Engineering Methodology to the Major Project (M.Tech)
    - Use of Microsoft Project as the software tool for managing their M.Tech project

Evaluation

Presentation of Systems Engineering Process of the Major Project

Textbook

1. Introduction to Systems Engineering by Ian Faulconbridge, Michael Ryan (Kindle Edition)
   Argos Press, 2015.
2. Systems Engineering Fundamentals
   https://ocw.mit.edu/courses/aeronautics...systems-engineering.../sefguide_01_01.pdf
3. https://www.sebokwiki.org/wiki/Introduction_to_Systems_Engineering

Web Resources

1. https://www.coursera.org/learn/systems-engineering
2. https://ocw.mit.edu/courses/aeronautics-and-astronautics/16-842-fundamentals-of-systems-engineering-fall-2015/

Title	**Innovation and IP Management** **Management of Innovation and IP**	Number	**OAL7XX0**
Department	OA	L-T-P [C]	1–0–0 [1]
Offered for	All M. Tech and PhD students <Shall also be considered for B.Tech Students>	Type	Compulsory
Prerequisite	Not required		

Objectives

This course is intended to familiarise students with basic concepts of IP, its relevance to innovations and their effective exploitation including emerging societal issues.

The Instructor will:

1. Highlight concept of Innovations and processes that leads to innovations.
2. Illustrate the significance of IPR in the context of Innovations and Demonstrate how information in IP data bases are used to enhance quality of innovations.
3. Explain how IP can be meaningfully Exploited in commerce and for benefit of society.

Learning Outcomes

The students are expected to have the ability to:

1. Use IP information as an effective resource for innovation.
2. Protecting innovations as IP.
3. Design IP strategies for protection of innovations and commercial exploitations.

Contents

Module I: Types of creations and innovations using the mind, Modes of IP protection of the output of the Human mind, Such as Patents, Design registrations, Trade Mark, Copyright, Trade Secrets, Geographical Indications. (5L)

Module-2 : Searching IP databases to extract technical information for research. Concept of Patentable inventions. Evaluating Inventions for patentability and drafting of patent specifications. Procedures for patent applications and their examination. Introducing processes for Design registrations, Trade Mark, Copyright, Trade Secrets, Geographical Indications. (6L)

Module-3 : Use of IP in Trade and Commerce. Societal issues. (3L)

Textbook